

Q2. Selection of a Linear Mixed Model for the Continuous Outcome

In Assignment 1, an exploratory analysis of the longitudinal BPRS measurements was conducted to investigate the mean structure, variability, and correlation patterns in the data. Three key features emerged: (i) a clear nonlinear time trend in BPRS scores, (ii) substantial between-patient heterogeneity in both baseline levels and longitudinal evolution, and (iii) strong within-patient dependence.

These findings motivate the use of a linear mixed-effects model (LMM) for the continuous outcome.

Model Specification

To model the longitudinal BPRS data, a linear mixed-effects model was specified with fixed effects for time, quadratic time, and baseline covariates (age, sex, BMI, ADL, amyloid PET, and tau PET). Between-patient heterogeneity and within-patient correlation were accounted for by including patient-specific random effects.

The general form of the model is:

$$\text{BPRS}_{ij} = \beta_0 + \beta_1 t_{ij} + \beta_2 t_{ij}^2 + \mathbf{x}_i^\top \boldsymbol{\beta} + b_{0i} + b_{1i} t_{ij} + \varepsilon_{ij}.$$

where b_{0i} and b_{1i} denote the random intercept and random slope for time, respectively.

Two competing random-effects structures were considered:

1. A model with a patient-specific random intercept only.
 2. A model with both a patient-specific random intercept and a random slope for time.
1. Both models were estimated using restricted maximum likelihood (REML). Model comparison was based on the -2 restricted log-likelihood, AIC, and BIC (Table 5).

Table 5. Comparison of model with patient-specific random intercept only (Model 1) and patient-specific random intercepts and a random slope for time (Model 2).

Fit Statistics	Model 1	Model 2
-2 Res LL	31278.9	29001.1
AIC	31282.9	29009.1
AICC	31282.9	29009.1
BIC	31293.2	29029.6

Model Selection

The comparison clearly favors the model including both a random intercept and a random slope for time. Introducing the random slope leads to a substantial improvement in fit, as evidenced by markedly lower values of -2 restricted log-likelihood, AIC, and BIC.

This indicates that patients differ not only in baseline BPRS levels but also in the rate of change over time. These results are consistent with the exploratory trajectory plots from Assignment 1, which showed pronounced heterogeneity in both intercepts and slopes. Consequently, the random-intercept-only model is inadequate, as it imposes a common time trend across individuals.

Final selected model

The final selected LMM therefore includes:

- fixed effects for time, quadratic time, and baseline covariates,
- a patient-specific random intercept, and
- a patient-specific random slope for time, with an unstructured covariance matrix for the random effects.

Residual errors were assumed to be homoscedastic, an assumption that was later examined in Question 5. Degrees of freedom for fixed-effect inference were adjusted using the Kenward–Roger method to improve small-sample accuracy.

Interpretation and Justification

The selected LMM provides a flexible yet parsimonious representation of the longitudinal data. By allowing patient-specific intercepts and slopes, the model captures heterogeneity in disease severity and progression, while preserving a clear interpretation of the population-average trajectory through the fixed effects.

Because the model is likelihood-based, inference remains valid under the Missing At Random assumption. This property is exploited in later questions addressing missingness and sensitivity. Overall, the selected model aligns closely with the exploratory findings from Assignment 1 and provides a principled foundation for subsequent analyses.